2.10.45

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Question

If $\mathbf{a}$ and $\mathbf{b}$ are two unit vectors such that $\mathbf{a} + 2\mathbf{b}$ and $5\mathbf{a} - 4\mathbf{b}$ are perpendicular to each other, then the angle between $\mathbf{a}$ and $\mathbf{b}$ is:

1. 45°
2. 60°
3. $\arccos \frac{1}{3}$
4. $\arccos \frac{2}{7}$

Step 1: The Perpendicularity Condition

The vectors are perpendicular, so their dot product is zero. In matrix notation:

$$(\mathbf{a} + 2\mathbf{b})^T (5\mathbf{a} - 4\mathbf{b}) = 0 \tag{1}$$

Expanding this expression:

$$\begin{aligned} (\mathbf{a}^T + 2\mathbf{b}^T)(5\mathbf{a} - 4\mathbf{b}) &= 0 \\ 5(\mathbf{a}^T\mathbf{a}) - 4(\mathbf{a}^T\mathbf{b}) + 10(\mathbf{b}^T\mathbf{a}) - 8(\mathbf{b}^T\mathbf{b}) &= 0 \end{aligned} \tag{2}$$

Step 2: Simplifying with Vector Properties

Since $\mathbf{a}^T\mathbf{b} = \mathbf{b}^T\mathbf{a}$ (dot product is commutative), we get:

$$5(\mathbf{a}^T\mathbf{a}) + 6(\mathbf{a}^T\mathbf{b}) - 8(\mathbf{b}^T\mathbf{b}) = 0 \tag{3}$$

The vectors are unit vectors, so their squared magnitude is 1:

$$\mathbf{a}^T\mathbf{a} = 1 \quad \text{and} \quad \mathbf{b}^T\mathbf{b} = 1 \tag{4}$$

Substituting this into equation (3):

$$5(1) + 6(\mathbf{a}^T\mathbf{b}) - 8(1) = 0 \implies \mathbf{a}^T\mathbf{b} = \frac{1}{2} \tag{5}$$

Step 3: Finding the Angle

The angle θ is defined by $\mathbf{a}^T\mathbf{b} = |\mathbf{a}||\mathbf{b}|\cos\theta$.

$$\cos\theta = \frac{\mathbf{a}^T\mathbf{b}}{|\mathbf{a}||\mathbf{b}|} = \frac{1/2}{(1)(1)} = \frac{1}{2} \tag{6}$$

Therefore, the angle is:

$$\theta = \arccos\left(\frac{1}{2}\right) = 60^\circ$$

The correct option is (b).

C Code - Verification

```
#include <stdio.h>
#include <math.h>

// Function to compute angle between two vectors
double find_angle(double a[2], double b[2]) {
    double dot = a[0]*b[0] + a[1]*b[1];
    double mag_a = sqrt(a[0]*a[0] + a[1]*a[1]);
    double mag_b = sqrt(b[0]*b[0] + b[1]*b[1]);
    return acos(dot / (mag_a * mag_b));
}

int main() {
    // Unit vectors with 60 deg between them
    double a[2] = {1, 0};
    double b[2] = {0.5, sqrt(3)/2};

    double theta = find_angle(a,b);
    printf(Angle = %.2f deg\n, theta*180/M_PI);
    return 0;
```

Python + C Code

```
import ctypes
import numpy as np

# Load C shared library
# Compile with: gcc -shared -o angle_vectors.so -fPIC angle.c -lm
lib = ctypes.CDLL(./angle_vectors.so)

# Define argtypes and restype
lib.find_angle.argtypes = [ctypes.POINTER(ctypes.c_double),
                           ctypes.POINTER(ctypes.c_double)]
lib.find_angle.restype = ctypes.c_double

# Define vectors
a = (ctypes.c_double * 2)(1, 0)
b = (ctypes.c_double * 2)(0.5, np.sqrt(3)/2)

# Call C function
theta = lib.find_angle(a, b) * 180 / np.pi
print(fAngle between vectors = {theta:.2f} degrees)
```

Pure Python Code

```python
import numpy as np
import matplotlib.pyplot as plt

# Define unit vectors
A = np.array([1, 0])
B = np.array([0.5, np.sqrt(3)/2])

# Compute angle
dot = np.dot(A, B)
norm_prod = np.linalg.norm(A) * np.linalg.norm(B)
theta = np.degrees(np.arccos(dot / norm_prod))
print(fAngle between vectors = {theta:.2f} degrees)
```

Python Plotting Code

```
# Plot vectors
O = np.array([0,0])
plt.quiver(*O, A[0], A[1], angles='xy', scale_units='xy', scale=1, color='r', label='a')
plt.quiver(*O, B[0], B[1], angles='xy', scale_units='xy', scale=1, color='b', label='b')

# Display angle
plt.text(0.2, 0.1, f{theta:.2f}, fontsize=12, color='purple')

plt.xlim(-0.2, 1.2); plt.ylim(-0.2, 1.2)
plt.grid(True); plt.gca().set_aspect('equal')
plt.title(Unit Vectors with 60 Angle)
plt.legend(); plt.savefig(figs/Figure_1.png)
plt.show()
```

Plot of the Unit Vectors